PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

# Exam 3 · Version B · Solutions

20 questions · 90 minutes · original practice exam, not an official exam

## ANSWER KEY

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **a** | 2 | **c** | 3 | **d** | 4 | **c** | 5 | **a** | 6 | **c** | 7 | **a** |
| 8 | **e** | 9 | **a** | 10 | **c** | 11 | **c** | 12 | **c** | 13 | **a** | 14 | **d** |
| 15 | **b** | 16 | **c** | 17 | **e** | 18 | **a** | 19 | **a** | 20 | **a** | | |

**THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.**

A door is 1.0 m wide. You push with a 40 N force at a point 0.50 m from the hinges, at 45° to the door.

**1)** What torque do you exert about the hinges?

**Answer: (a) 14.1 N·m**

$\tau = rF\sin\varphi = (0.5)(40)(\sin 45°) = 14.1$ N·m.

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**THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

An Atwood machine has masses of 3.0 kg and 5.0 kg on either side of a pulley that is a uniform solid disk (M = 4.0 kg, R = 0.15 m). The rope does not slip and the axle is frictionless.

**2)** What is the magnitude of the acceleration of the masses?

**Answer: (c) 1.96 m/s²**

$a = (m_2 - m_1)g/(m_1 + m_2 + M/2) = (2)(9.81)/10 = 1.96$ m/s².

**3)** What is the tension in the rope on the 3.0 kg side?

**Answer: (d) 35.3 N**

$T_1 = m_1(g + a) = 3(9.81 + 1.96) = 35.3$ N.

**4)** How do the tensions on the two sides compare?

**Answer: (c) The tension on the 5.0 kg side is larger.**

A net torque is needed to give the massive pulley its angular acceleration, so $T_2 > T_1$.

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**THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A thin hoop and a solid cylinder, with the same mass and radius, are released together from rest at the top of the same incline and roll without slipping. The top is 2.0 m above the bottom.

**5)** Which reaches the bottom first?

**Answer: (a) The solid cylinder**

The cylinder has a smaller $I/MR^2$, so less of its energy goes into rotation.

**6)** What is the speed of the hoop at the bottom?

**Answer: (c) 4.43 m/s**

$mgh = \frac{1}{2}mv^2 + \frac{1}{2}(mR^2)(v/R)^2 = mv^2 \rightarrow v = \sqrt{(gh)} = 4.43$ m/s.

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**THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A grinding wheel is a uniform disk of mass 5.0 kg and radius 0.20 m, spinning at 300 rad/s. A tool is pressed against its rim with a normal force of 40 N, and the coefficient of kinetic friction is 0.60.

**7)** What is the magnitude of the angular acceleration of the wheel?

**Answer: (a) 48 rad/s²**

$\tau = \mu N \cdot R = (0.6)(40)(0.2) = 4.8$ N·m; $I = \frac{1}{2}MR^2 = 0.1$ kg·m²; $\alpha = \tau/I = 48$ rad/s².

**8)** How long does the wheel take to stop?

**Answer: (e) 6.25 s**

$t = \omega/\alpha = 300/48 = 6.25$ s.

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**THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A uniform horizontal beam 4.0 m long with a mass of 20 kg is hinged to a wall at one end. A cable attached to the other end makes a 30° angle with the beam. A 50 kg sign hangs from the end of the beam.

**9)** What is the tension in the cable?

**Answer: (a) 1180 N**

Torques about the hinge: $T \sin 30°(4) = (20g)(2) + (50g)(4)$, $T = 1180$ N.

**10)** What is the horizontal force the hinge exerts on the beam?

**Answer: (c) 1020 N**

$\Sigma F_x = 0$: $H = T \cos 30° = 1020$ N.

**11)** If the cable were attached to the wall higher up, so it made a larger angle with the beam, the tension would

**Answer: (c) decrease.**

A larger angle gives a larger perpendicular component, so less tension is needed for the same torque.

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**THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

Disk A ($I = 0.50$ kg·m²) spins at 30 rad/s. Disk B ($I = 1.0$ kg·m²), initially not spinning, is dropped onto it along the same axle, and the disks stick together.

**12)** What is the final angular speed?

**Answer: (c) 10 rad/s**

$I_A\omega_A = (I_A + I_B)\omega$, $\omega = (0.5)(30)/1.5 = 10$ rad/s.

**13)** What fraction of the initial kinetic energy is lost?

**Answer: (a) 66.7%**

$K_f/K_i = I_A/(I_A + I_B) = 1/3$, so 2/3 is lost.

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**THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A 2.0 kg ball moves at a constant 5.0 m/s in the +x direction along the line y = 3.0 m.

**14)** What is the magnitude of the ball's angular momentum about the origin?

**Answer: (d) 30 kg·m²/s**

$L = mvr_\perp = (2)(5)(3) = 30$ kg·m²/s.

**15)** As the ball moves along the line, its angular momentum about the origin

**Answer: (b) stays constant.**

No force acts, so no torque acts, and the perpendicular distance stays 3.0 m.

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**THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A 0.50 kg block on a spring moves according to $x(t) = (0.12\text{ m})\cos(8.0t)$, with t in seconds.

**16)** What is the spring constant?

**Answer: (c) 32 N/m**

$\omega^2 = k/m$, so $k = (0.5)(8^2) = 32$ N/m.

**17)** What is the total mechanical energy of the oscillation?

**Answer: (e) 0.23 J**

$E = \frac{1}{2}kA^2 = \frac{1}{2}(32)(0.12^2) = 0.23$ J.

**18)** If the mass were four times larger (same spring), the period would

**Answer: (a) double.**

$T = 2\pi\sqrt{(m/k)} \propto \sqrt{m}$.

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**THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.**

A 0.30 kg mass hangs at rest from a vertical spring, which it stretches by 6.0 cm.

**19)** If the mass is pulled down slightly and released, what is the period of oscillation?

**Answer: (a) 0.491 s**

$k = mg/\Delta y = 49.1$ N/m; $T = 2\pi\sqrt{(m/k)} = 2\pi\sqrt{(\Delta y/g)} = 0.491$ s.

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**THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.**

You want to build a simple pendulum with a period of 1.5 s on Earth.

**20)** How long should the pendulum be?

**Answer: (a) 0.559 m**

$L = g(T/2\pi)^2 = 9.81(1.5/2\pi)^2 = 0.559$ m.

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